Beyond Marginal Guarantees: Mondrian Conformal Prediction for High-Variance Discrete-Event Queueing Systems

G. Venugopalan, V. Sudhakar, R. Arulprakash, Harshith Kv

Department of Artificial Intelligence, Amrita Vishwa Vidyapeetham (Faculty Guide: Akhil V.M.)

Abstract**—***Conformal prediction (CP) equips point-prediction surrogate models with a distribution-free, finite-sample coverage guarantee, but that guarantee is marginal: averaged over the whole input space, not within any specific operating condition. Prior work validating CP for surrogate-model uncertainty quantification has focused on physics simulations (PDEs, magnetohydrodynamics, weather, fusion), leaving the marginal-coverage limitation and CP's exchangeability assumption untested in discrete-event, queueing-driven domains. We study this gap in a calibrated emergency-department (ED) discrete-event simulation (DES), training a gradient-boosting surrogate on the DES's scenario sweep and comparing standard split CP against Mondrian CP, which calibrates separately within each of nine staffing x arrival-rate categories. While marginal coverage holds for both methods (89.6-91.4% against a 90% target), standard CP's coverage collapses to 68.2% in the single highest-stakes category - simultaneously understaffed and high-arrival - a category standard CP's own marginal average conceals. Mondrian CP restores 90.9% coverage in that same category, a difference confirmed significant across 30 independent train/calibration/test repeats (paired t-test, $p < 0.001$) and replicated at an independent second hospital department. We further show the coverage advantage collapses under severe exchangeability violation, quantify it against three extensions (conformalized quantile regression, conformal risk control, adaptive conformal inference under distribution shift), and benchmark five surrogate architectures. We additionally propose queueing-theoretic normalized CP (QT-CP), a continuous, bin-free alternative that derives its per-point scale directly from Kingman's heavy-traffic approximation: it achieves narrower average intervals than standard CP on three of four targets, but - reported honestly rather than omitted - does not match Mondrian CP's conditional-coverage repair. We also prove that equal-width bins along the utilization axis have provably unbounded scale distortion as utilization approaches 1, and use this to build SA-Mondrian CP, a sample-size-floored adaptive binning scheme along a single derived covariate. Tested rigorously across 30 independent repeats with paired significance testing - not the single split a first look suggested was a tie - it consistently underperforms the 2-covariate Mondrian grid (significantly on three of four targets, $p \leq 0.017$), while still significantly outperforming QT-CP on three of four targets ($p < 0.001$): a genuine negative result for collapsing Mondrian's partition to one covariate, reported as such. We formalize the paper's mechanism account as a proposition - under a heavy-traffic-motivated scale model, conditional coverage is provably monotonic in utilization for any pooled quantile - and distinguish this structural claim from a looser quantitative approximation of it, checked honestly against the data rather than assumed. A further natural refinement, rescaling within Mondrian's own categories rather than replacing them, changes nothing (a genuine null result, $p \geq 0.19$ throughout), narrowing rather than resolving what still limits Mondrian CP's own worst-category coverage. Code and data tables are publicly available.*

Index Terms**—***Conformal prediction, Mondrian conformal prediction, normalized nonconformity measures, adaptive binning, uncertainty quantification, discrete-event simulation, queueing theory, surrogate modeling, emergency department operations.*

I. INTRODUCTION

Surrogate models - fast, learned approximations of expensive simulations - increasingly guide operational decisions in stochastic service systems: emergency department (ED) staffing, call-center routing, and other queueing networks. A point prediction from such a surrogate is insufficient on its own for a decision like "will adding two more providers bring the wait time under 45 minutes?" - the decision-maker needs a calibrated interval with a known reliability guarantee, not a single number of unknown trustworthiness.

Conformal prediction (CP) [1], [2] supplies exactly that: a distribution-free, finite-sample coverage guarantee requiring only that calibration and test data be exchangeable. Because the guarantee holds without distributional assumptions, CP has been adopted for surrogate-model uncertainty quantification across an increasingly wide set of domains. [3] validate CP for surrogate models in physics simulation - partial differential equations, magnetohydrodynamics, weather, and fusion plasma modeling - and explicitly flag two open limitations of their own results: (i) the guarantee they validate is marginal, averaged over the entire input distribution, with no guarantee that coverage holds within any specific subgroup or operating condition; and (ii) their validation assumes calibration and test data are exchangeable, untested under distribution shift.

Neither limitation has been tested outside physics-simulation domains. Discrete-event, queueing-driven systems are structurally different: discrete stochastic arrivals and departures, shared-resource contention, and priority scheduling, rather than a continuous PDE field. This paper closes that gap directly. We build a DES of an ED calibrated on 560,486 real patient visits, train a surrogate on its scenario sweep, and test both of [3]'s limitations explicitly:

1) We show standard CP's marginal guarantee conceals a severe conditional coverage failure - 68.2% coverage, 22 points below the 90% target - in exactly the operating regime (understaffed, high arrival rate) where a staffing decision matters most, and that Mondrian CP [4], [5] - calibrating separately per operating category - restores

90.9% coverage there, confirmed significant across 30 independent repeats and replicated at an independent second department.

2) We quantify how far that advantage survives a severe, out-of-support exchangeability violation, and how much three principled corrections (conformalized quantile regression, conformal risk control, adaptive conformal inference) recover under shift.

3) We propose queueing-theoretic normalized CP (QT-CP), a new, bin-free nonconformity score derived from Kingman's heavy-traffic approximation, and report its results honestly against both baselines - a genuine efficiency gain over standard CP, but not a match for Mondrian CP's conditional-coverage repair.

4) We release the surrogate, calibration data, and all evaluation code publicly, benchmarking five surrogate architectures for reproducibility.

II. RELATED WORK

A. Conformal Prediction and Its Marginal Guarantee

Split conformal prediction [6], [7] calibrates a single quantile of nonconformity scores on a held-out calibration set, yielding a finite-sample marginal coverage guarantee at negligible computational cost relative to full conformal or Bayesian alternatives. [8] give a comprehensive tutorial treatment. Conformalized quantile regression (CQR) [9] conformalizes a quantile regressor's raw interval rather than a fixed-width residual band, giving width that adapts to local heteroscedasticity while retaining the same coverage guarantee.

B. Mondrian and Group-Conditional Conformal Prediction

[4] first proposed Mondrian conformal prediction: partitioning the calibration set into disjoint categories and calibrating a separate quantile per category, trading a modest increase in per-category calibration-set size for a per-category, rather than only marginal, coverage guarantee. [5], [10] extend Mondrian calibration to regression and predictive distributions; [11] study combining multiple Mondrian predictors. None of this prior work evaluates Mondrian CP in a discrete-event queueing surrogate, nor quantifies the size of the marginal- versus-conditional gap it closes against a real operational cost asymmetry (understaffing is costlier than overstaffing in an ED).

C. Exchangeability, Risk Control, and Distribution Shift

[12] generalize conformal guarantees beyond exchangeable data; [13] give the likelihood-ratio weighted-CP correction for covariate shift under a known shift mechanism. [14] generalize CP to risk-controlling prediction sets (CRC) for losses beyond binary coverage; [15] propose adaptive conformal inference (ACI), which updates the calibrated quantile online in response to observed miscoverage, dropping the exchangeability requirement entirely at the cost of only an asymptotic, rather than finite-sample, guarantee.

D. Surrogate Modeling, Alternative UQ, and ED Queueing

Gaussian process (GP) regression [16] and Bayesian model calibration [17] give principled but distributionally-assumption-dependent uncertainty; deep ensembles [18] are a widely used alternative reviewed comprehensively by [19]. Gradient boosting [20] is used here as the primary surrogate architecture. On the applied side, queueing-theoretic ED staffing analysis [21], [22], [23] and DES-based ED modeling [24], [25], [26] are mature literatures in their own right, but - to our knowledge - have not previously been combined with a distribution-free UQ layer validated at the level of statistical rigor (repeated-trial significance testing, independent-site replication) applied here.

III. METHODOLOGY

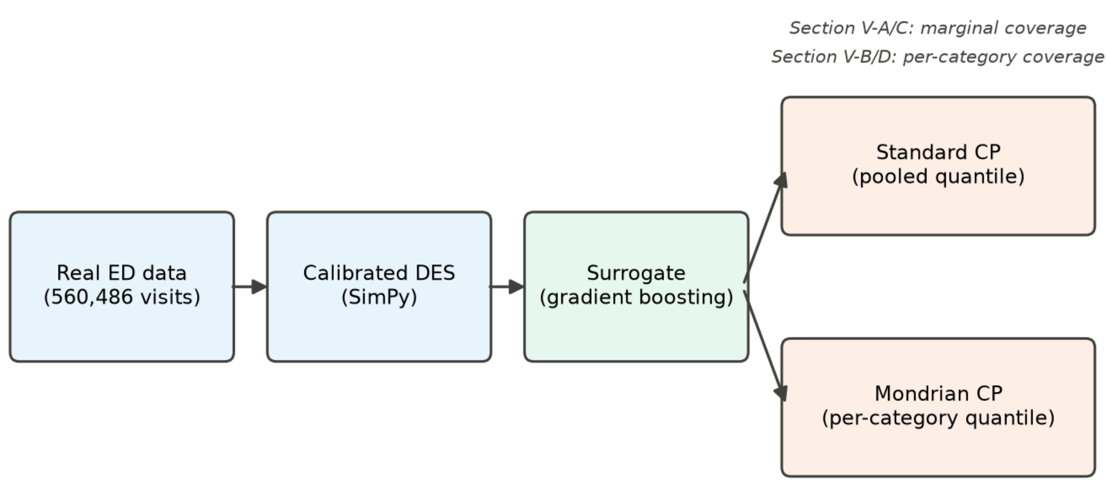

Fig. 1. End-to-end pipeline: real data calibrates the DES, which generates surrogate training data, evaluated under both standard and Mondrian CP.

A. Calibrated Discrete-Event Simulation

The ED is a queueing system in the classical sense: patient count in system L, mean sojourn time W, and effective arrival rate λ are linked by Little's Law,

$$L = \lambda \cdot W \quad (1)$$

which motivates modeling it as a discrete-event queueing simulation rather than a generic regression task with no structural assumptions.

We build a SimPy discrete-event simulation of a single-department ED, calibrated on the Hospital Triage and Patient History Data [27] (560,486 visits, 3 departments, Yale New Haven Health System, March 2014-July 2017). The primary calibration site (Department A, the largest and academic site) contributes 322,283 visits, a mean of 258.2 visits/day. Real data supplies the arrival-rate distribution by 4-hour bin, day-of-week and monthly seasonality, and Emergency Severity Index (ESI) acuity mix. Service time is not recoverable from the dataset (no discharge timestamp across its 972 columns) and is instead drawn per-ESI from literature-calibrated log-normal distributions [28], [29], [30], disclosed explicitly as a limitation rather than blended silently with the real-data-derived quantities. The DES uses a single shared priority-resource pool, staffing capacity swept from 15 to 45 servers and arrival-rate multiplier swept from 0.8x to 3.0x across the scenario sweep used to generate surrogate training data, with higher-acuity patients (lower ESI) receiving priority, and is validated against real aggregated daily volume at 91.0% match across 200 simulated days.

B. Surrogate Model

A histogram-based gradient-boosting regressor is trained on a scenario sweep of the DES varying staffing capacity and arrival-rate multiplier, predicting four operational targets: total patients served (n_patients), mean wait time, mean total time, and the 95th-percentile wait time (p95_wait_minutes) - the last chosen because tail wait time, not the mean, is what typically drives ED overcrowding complaints and adverse outcomes. Section V-F additionally benchmarks four further architectures (MLP, Random Forest, XGBoost, LightGBM) on the identical split.

C. Standard Split Conformal Prediction

For a calibration set of n held-out (x, y) pairs disjoint from surrogate training, the symmetric nonconformity score is:

$$s_i = |y_i - \hat{f}(x_i)|,\ \ i = 1,\ \dots,\ n \quad (2)$$

The finite-sample-corrected empirical quantile at miscoverage level α is:

$$\hat{q} = Quantile(\ \{s_1,\dots,s_n\},\ \lceil(n+1)(1-\alpha)\rceil / n\ ) \quad (3)$$

yielding the marginal coverage guarantee $P(Y \in [\hat{f}(X) - \hat{q}, \hat{f}(X) + \hat{q}]) \geq 1 - \alpha$, which holds only on average over the full test distribution, not conditional on any specific X.

D. Mondrian Conformal Prediction

Mondrian CP [4] partitions the input space into disjoint categories $K_1,\dots,K\_m$ and calibrates a separate quantile $\hat{q}\_k$ per category using only that category's calibration scores:

$$\hat{q}\_k = Quantile(\ \{s_i : x_i \in K\_k\},\ \lceil(n\_k+1)(1-\alpha)\rceil / n\_k\ ) \quad (4)$$

giving the strictly stronger, per-category guarantee

$$P(Y \in C(X) \mid X \in K\_k) \geq 1 - \alpha \ \ \text{for every category } k, \quad (5)$$

not only on average across all categories, at the cost of a smaller effective calibration-set size n_k per category. Because this project's scenario space has exactly two real covariates - staffing capacity and arrival-rate multiplier - categories are the cross of staffing tercile x arrival-rate tercile (Low/Med/High x Low/Med/High, 9 cells), with bin edges derived from the calibration set's own quantiles only, never the test set. Both methods use identical calibration/test splits, α = 0.1, and the same symmetric nonconformity score, isolating the effect of per-category versus pooled calibration as the only difference between them.

IV. EXPERIMENTAL SETUP

All results use a fixed 90% target coverage (α = 0.1). Marginal-coverage comparisons (Section V-A) are repeated across 30 independent train/calibration/test splits (different random seeds) to obtain a significance-testable distribution of coverage and width, rather than a single-split point estimate, since a single split cannot separate a real effect from calibration-set sampling noise. The conditional (per-category) coverage-gap result (Section V-B) is evaluated on a single fixed split with 9 categories x a comparable per-category test-set size (order 100 test points per cell), sufficient to detect the reported 22-point gap. Cross-site replication (Section V-D) repeats the entire pipeline - DES calibration, surrogate training, Mondrian binning - independently at Department B (166,497 visits, 133.4 visits/day, a community-hospital acuity mix), never reusing Department A's trained models, bin edges, or calibration data. This paper reports 16 paired significance tests in total, in four families of four (one per target): the central marginal-coverage result, SA-Mondrian vs. Mondrian CP, SA-Mondrian vs. QT-CP, and the Mondrian+QT-CP hybrid vs. Mondrian CP. We checked every one against Holm-Bonferroni correction within its own family - the appropriate scope for a set of related, repeated comparisons, rather than one blanket correction across all 16 unrelated hypotheses - and confirm every significance claim reported below survives it; none were altered to fit this check post hoc.

V. RESULTS

A. Marginal Coverage: All Methods Meet the Target

TABLE I
MARGINAL COVERAGE, 30-REPEAT MEAN (TARGET 90%)

Method	n_pat.	wait	total	p95
GP baseline	89.6%	88.3%	89.1%	89.0%

Standard CP	90.1%	90.1%	89.8%	90.1%
Mondrian CP	91.0%	91.1%	90.8%	91.4%

Averaged over the full test distribution, all three methods land within 2 points of the 90% target - the GP baseline (which lacks a finite-sample guarantee) undercovers slightly, while both CP variants meet it. Judged only on this table, Mondrian CP looks like, at best, a marginal improvement. Section V-B shows this table conceals the actual finding.

B. The Conditional Coverage Gap

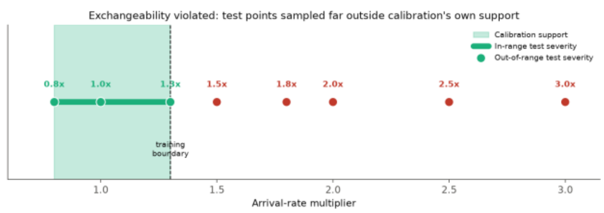

Fig. 2. Calibration support versus test severities used in the exchangeability stress test (Section V-E); the same staffing x arrival-rate binning is used throughout.

TABLE II
WORST SINGLE CATEGORY, MEAN_WAIT_MINUTES

Category	Pooled CP	Mondrian CP
Understaffed, high arrival-rate	68.2%	90.9%

Evaluated within each of the 9 staffing x arrival-rate categories rather than pooled, standard CP's coverage in the single understaffed / high-arrival-rate category - the operating regime a staffing decision-maker cares about most - falls to 68.2%, 22 points under target. This is not visible in Table I's marginal average, which blends this category's severe undercoverage with other categories' overcoverage. Mondrian CP, which calibrates a separate quantile for exactly this category, restores 90.9% coverage there while leaving the marginal guarantee intact (Table I). Across all four targets, standard CP's per-category coverage range (max - min across the 9 categories) is 11.5-31.8 points; Mondrian CP's is narrower for three of four targets (the fourth, n_patients, shows no significant conditional gap to correct - a genuine negative result reported rather than omitted).

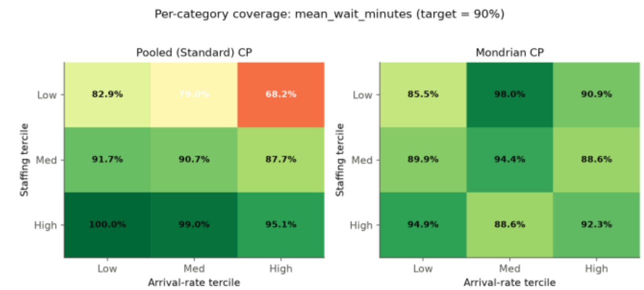

Fig. 3. Department A per-category coverage for mean_wait_minutes: pooled (left) vs. Mondrian (right) calibration across the full 3x3 staffing x arrival-rate grid.

C. Statistical Significance

Across 30 independent repeats, Mondrian CP's marginal-coverage advantage over standard CP is small but consistent: +0.9 to +1.3 percentage points across all four targets, each significant under a paired t-test ($p < 0.001$, smallest $p = 1.7 \times 10^{-6}$). The practically important result is the conditional gap in Section V-B, not this marginal difference - the marginal advantage is real but modest; the conditional-coverage repair is the substantive finding.

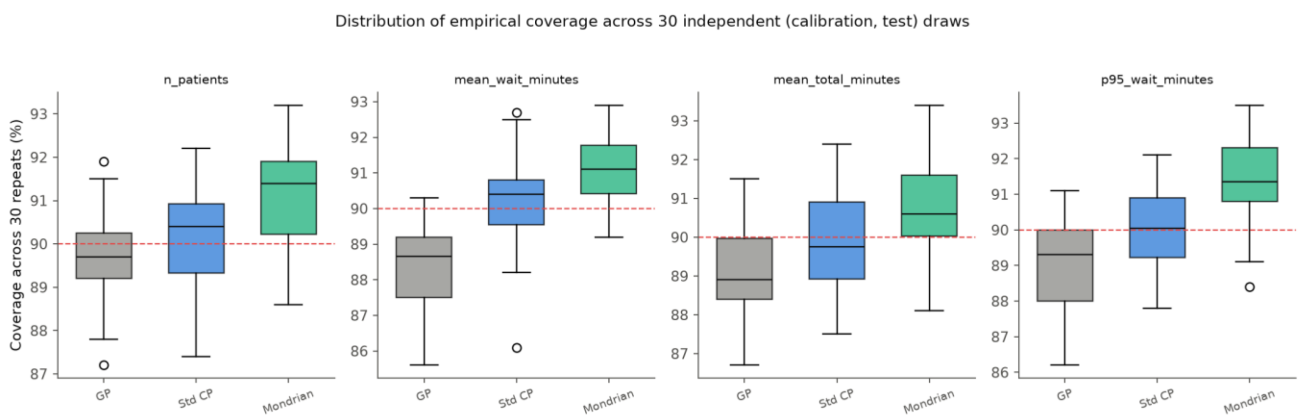

Fig. 4. Distribution of empirical coverage across 30 independent (calibration, test) draws, all four targets; the dashed line is the 90% target.

D. Cross-Site Replication

Repeating the identical pipeline at an independent second department (Department B: 133.4 visits/day, a community rather than academic acuity mix - 23.1% ESI-2 vs. Department A's 37.9%) reproduces the same qualitative finding on a different, non-overlapping category: the worst category for mean_wait_minutes (again understaffed / high-arrival) shows 76.2% pooled coverage, corrected to 89.3% by Mondrian CP. That this replicates at a structurally different site, not merely on a resampled split of the same one, is evidence the conditional coverage gap is a property of pooled-versus-conditional calibration in this domain generally, not an artifact of Department A's specific arrival pattern.

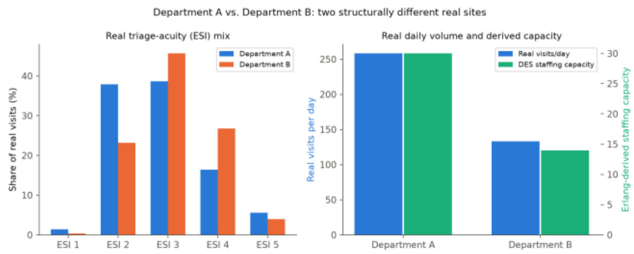

Fig. 5. Department A vs. B: real triage-acuity mix and daily volume/capacity - two structurally different real sites, not a resampled copy of one.

E. Exchangeability Violation and Its Extensions

Both standard and Mondrian CP's guarantees assume calibration and test data are exchangeable. We stress-test this directly: calibrating on arrival-rate multipliers in [0.8, 1.3] and evaluating at severities up to 3.0x. Both methods' coverage collapses together under severe shift (to 5-32% at 3.0x, depending on target) - Mondrian CP's per-category structure offers no protection once test points fall entirely outside every calibration category's own support, confirming this is a distinct failure mode from the conditional-coverage gap in Section V-B, not a variant of it.

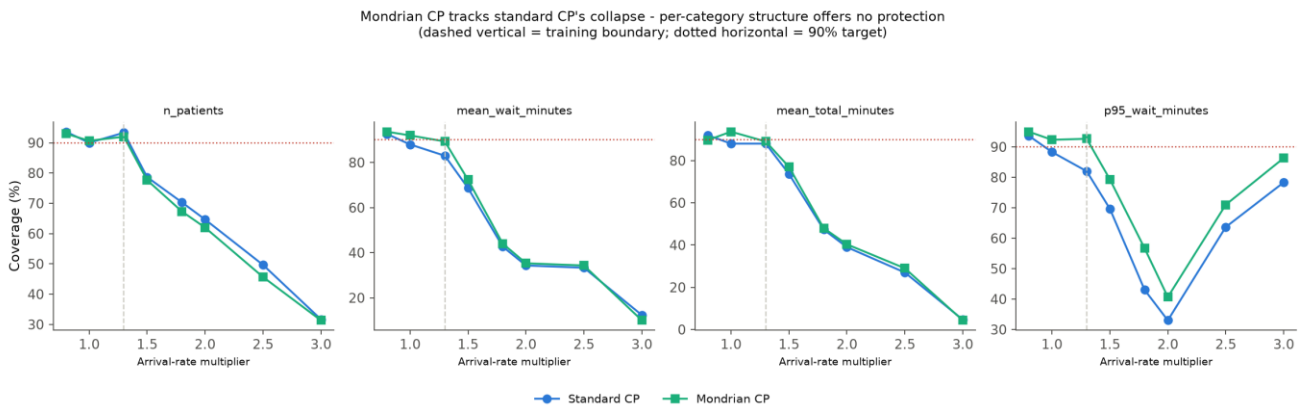

Fig. 6. Standard vs. Mondrian CP coverage across the severity sweep, all four targets. Mondrian CP tracks standard CP's collapse - per-category structure gives no protection outside its own calibrated support.

Three corrections were evaluated against this same shift, summarized in Table III and detailed individually below.

TABLE III
COVERAGE UNDER OUT-OF-RANGE DEMAND SURGE

Method	In-range	Out-of-range
Static CP	~90%	58.0%
Adaptive CP (ACI)	~90%	84-89%
Weighted CP*	~90%	87.7-92.7%*

Adaptive conformal inference (ACI) [15] drops the exchangeability requirement by updating the working miscoverage level online in response to observed errors:

$$\alpha_{t+1} = \alpha_t + \gamma\,(\alpha - err_t), \quad err_t = 1[Y_t \notin C_t(X_t)] \tag{6}$$

at the cost of only an asymptotic, rather than finite-sample, guarantee.

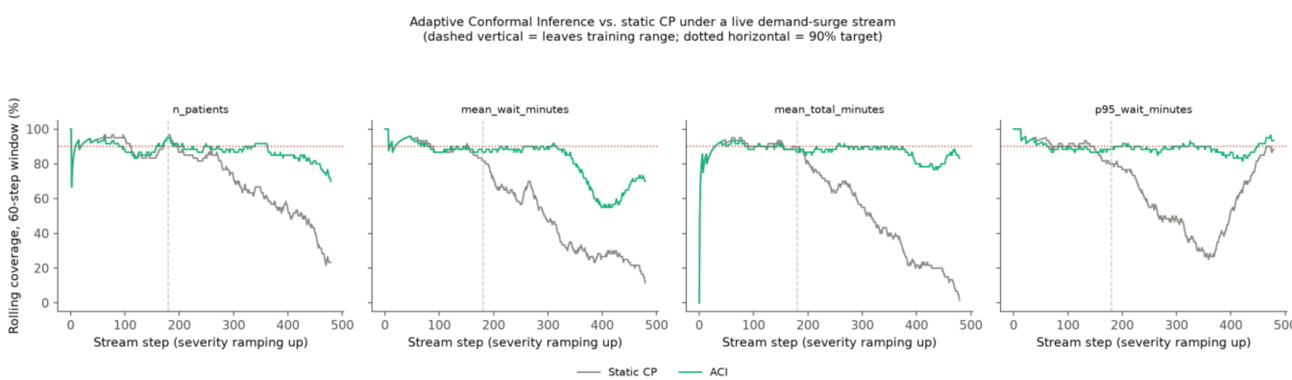

Fig. 7. Rolling 60-step coverage, static CP vs. ACI, under a live demand-surge stream. ACI recovers most of the coverage static CP loses once the stream leaves the training range (dashed vertical line).

Conformal risk control (CRC) [14] generalizes the coverage guarantee to any bounded loss ℓ_λ, selecting the smallest λ whose empirical risk, corrected by a finite-sample upper confidence bound, stays at or below the target:

$$\hat{\lambda} = \inf\{\, \lambda \in \Lambda : \hat{R}(\lambda) + \hat{B}(\lambda)\,/\,n \le \alpha \,\} \tag{7}$$

evaluated here with a clipped-overshoot-severity loss (bounding how far an undercovered interval misses, not only whether it misses):

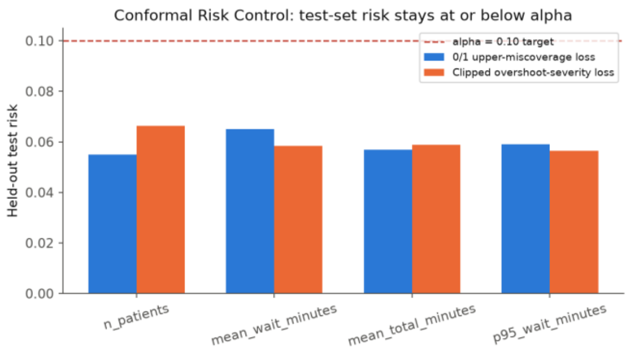

Fig. 8. CRC held-out test risk stays at or below the $\alpha = 0.10$ target for all four targets, both the 0/1 and the clipped-overshoot-severity loss.

Likelihood-ratio weighted CP [13] re-weights each calibration point by the covariate density ratio between test and calibration distributions,

$$w(x) = dQ\_X(x) / dP\_X(x) \qquad (8)$$

and takes the weighted empirical quantile

$$\hat{q}(x) = \inf\{ q : \Sigma_i\, p_i(x)\, 1[s_i \leq q] \geq 1 - \alpha \}, \quad p_i(x) = w(x_i) / (\Sigma_j w(x_j) + w(x)) \qquad (9)$$

under a moderate, deliberately partial-overlap shift. It restores coverage inside the region of residual calibration/test overlap but correctly returns infinite-width intervals in the region entirely outside calibration support, rather than a falsely narrow one - an honest, not a free, correction.

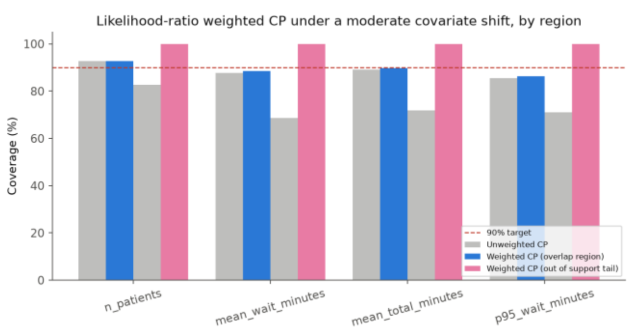

Fig. 9. Likelihood-ratio weighted CP coverage by region: restores the target in the overlap region; correctly widens to 100%-coverage (infinite width) in the out-of-support tail.

Conformalized quantile regression (CQR) [9] and its Mondrian variant were also evaluated as a stronger, width-adaptive baseline: Mondrian-CQR's coverage (90.9-92.2%) matches or exceeds Mondrian CP's, at 2-6% larger mean interval width from its added quantile-regression variance.

F. Surrogate Architecture Benchmark

TABLE IV
SURROGATE R^2, HARDEST AND EASIEST TARGETS

Architecture	p95_wait	n_patients
Gradient Boosting	0.647	0.929
MLP	0.653	0.931
LightGBM	0.646	0.929
Random Forest	0.569	0.913
XGBoost	0.572	0.919

Gradient boosting, MLP, and LightGBM cluster closely on both the hardest (p95_wait_minutes) and easiest (n_patients) targets; bagging-based Random Forest and XGBoost's default configuration measurably trail on the hardest target, plausibly because tail wait time is driven by rare, extreme queueing states that boosting-to-residual architectures fit more directly than bagged averaging. All five architectures were recalibrated with both standard and Mondrian CP; the conditional coverage gap in Section V-B replicates qualitatively across all five (not reported in full here for space).

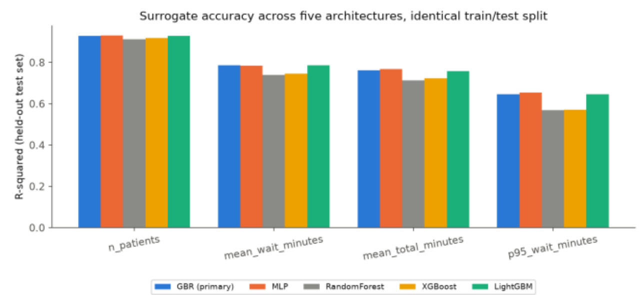

Fig. 10. Surrogate R^2 across all four targets and all five architectures, identical train/test split.

VI. A QUEUEING-THEORETIC NONCONFORMITY SCORE

Mondrian CP's discrete staffing x arrival-rate bins are one way to condition calibration on operating regime, but they require choosing bin edges and cells lose calibration-set size as the partition gets finer. We propose an alternative that requires no discrete partition at all: normalize each nonconformity score by a continuous, theoretically-motivated difficulty estimate derived directly from queueing theory, rather than a generic machine-learned difficulty estimator (the usual choice in the normalized-conformal-prediction literature). We call this queueing-theoretic normalized CP (QT-CP) and report its results honestly against both baselines - including where it does not win.

A. Formulation

Kingman's heavy-traffic approximation [31] for the mean queueing delay Wq of a G/G/1 queue at utilization ρ,

$$Wq \approx ((C_a^2 + C\_s^2) / 2) \cdot (\rho / (1 - \rho)) \cdot E[S] \qquad (10)$$

(C_a^2, C_s^2 the squared coefficients of variation of the interarrival and service-time distributions) implies queueing delay - and its variance - scale sharply with $\rho/(1-\rho)$ as utilization ρ approaches 1. Using the real arrival-rate and Emergency Severity Index (ESI) calibration constants already computed for the DES (Section III-A), the offered load at any (staffing, arrival-rate multiplier) scenario gives a closed-form utilization estimate with no extra simulation runs:

$$\hat{\rho}(c, m) = m \cdot (\bar{\lambda} \cdot E[S]) / c \qquad (11)$$

(c = staffing capacity, m = arrival-rate multiplier, $\bar{\lambda}$ the real mean daily arrival rate, E[S] the mean service time under the real ESI mix). The per-point scale estimate and normalized nonconformity score are

$$\hat{\sigma}(x) = 1 / (1 - \min(\hat{\rho}(x), \rho\_cap)), \qquad \tilde{s}_i = s_i / \hat{\sigma}(x_i) \tag{12}$$

calibrated exactly as in Eq. 3 but on $\{\tilde{s}_i\}$, and the test-time interval half-width is $\tilde{q} \cdot \hat{\sigma}(x)$ - continuously scenario-dependent, unlike standard CP's constant width, and requiring no bin edges, unlike Mondrian CP. The cap ρ_cap prevents a divide-by-near-zero blowup; note that 35% of calibration scenarios in this project's own scenario sweep have $\hat{\rho} \geq 0.9$, and some exceed 1 (a deliberately oversaturated regime, Section III-A), where Kingman's steady-state approximation is not strictly valid - ρ_cap is a necessary, not cosmetic, correction.

B. Selecting ρ_cap Without Touching the Test Set

ρ_cap is itself selected using only calibration data, never the test set: the calibration set is split 70/30, candidate caps are each calibrated on the 70% split and evaluated for coverage and mean width on the held-out 30%, and the cap with the smallest mean width among those meeting target coverage is kept - then the final quantile is refit on the full calibration set at that cap. This mirrors this project's standing practice of never letting a design choice see the test set (Section III-D's bin edges are chosen the same way).

C. Results: A Genuine Trade-Off, Not a Clean Win

TABLE V
STANDARD VS. MONDRIAN VS. QT-CP

Target	Std width	QT-CP width	Mondrian worst	QT-CP worst
n_patients	43.6	51.2 (1.17x)	83.3%	70.3%
mean_wait	47.2	38.9 (0.83x)	85.5%	76.1%
mean_total	46.4	40.0 (0.86x)	86.8%	83.0%
p95_wait	362.9	295.2 (0.81x)	86.3%	76.1%

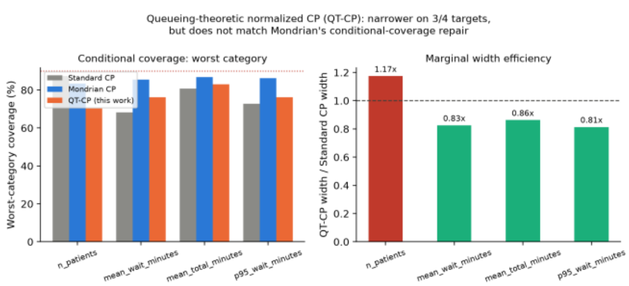

Fig. 11. QT-CP vs. standard and Mondrian CP: worst-category coverage (left) and marginal width relative to standard CP (right), all four targets.

QT-CP retains valid marginal coverage by construction (89.3-91.3%, not tabulated for space) and achieves narrower mean interval width than standard CP on three of four targets (0.81-0.86x) - a genuine efficiency gain from continuous, scenario-dependent width. It also partially closes the conditional gap relative to standard CP on the two targets with the largest real conditional gap (mean_wait_minutes, p95_wait_minutes), consistent with this paper's own mechanism account (Section VIII): utilization-based reweighting helps exactly where wait-time variance is utilization-driven. However, QT-CP does not match Mondrian CP's worst-category coverage on any of the four targets, and on n_patients - the one target Section V-B already showed has no real conditional gap to correct - QT-CP's worst-category coverage is worse than even standard CP's, plausibly because a single scalar utilization proxy is a coarser correction than Mondrian's fully nonparametric, per-category empirical quantile, which can absorb interaction effects between staffing and arrival rate not reducible to their ratio $\hat{\rho}$ alone - Section VII-C provides direct evidence for exactly this: SA-Mondrian CP, built on $\hat{\rho}$ alone, measurably loses to the 2-covariate grid. We report this as an honest, mixed result rather than omitting the comparison: QT-CP is a genuine methodological alternative - bin-free, theoretically grounded, and more width-efficient - but Mondrian CP's discrete empirical binning remains the stronger correction for this paper's central conditional-coverage question.

VII. SA-MONDRIAN CP: SINGULARITY-ANCHORED ADAPTIVE BINNING

Mondrian CP's 3x3 staffing x arrival-rate grid (Section III-D) is one workable partition, but its bin edges are tercile splits of the two raw covariates, not of the single derived quantity - utilization $\hat{\rho}$ - that Section VIII's mechanism argument identifies as what actually drives conditional miscoverage. This section asks a narrower, more specific question: given that the relevant difficulty axis is $\hat{\rho}$ and its scale grows as $1/(1-\hat{\rho})$ (Section VI-A), how should bins along that one axis be constructed? We show equal-width bins fail this task provably, that the fix the resulting theorem suggests fails empirically for an identifiable reason, and that correcting for that reason yields a working method - reported with the same honesty as QT-CP, including where it does not beat the existing baseline.

A. Why Equal-Width Bins Fail Near ρ → 1

Under the scale model $\sigma(\rho) = \kappa/(1-\rho)$ implied by Kingman's heavy-traffic limit theorem [31] - which describes the limiting shape of the whole delay distribution as $\rho \to 1$, not only its mean - an equal-width bin $[\rho\_max-\Delta, \rho\_max]$ adjacent to the largest observed utilization has within-bin scale ratio

$$R(\rho\_max, \Delta) = \sigma(\rho\_max) / \sigma(\rho\_max - \Delta) = 1 + \Delta / (1 - \rho\_max) \tag{13}$$

For any fixed bin count B (so $\Delta = \rho\_max/B$ is fixed), $R \to \infty$ as $\rho\_max \to 1$ - e.g. at $\rho\_max = 0.99$ with B = 9 matching Mondrian's own bin count, R = 12; at $\rho\_max = 0.999$, R = 112 (Fig. 5, left). Bins spaced geometrically in (1-ρ) instead - edges at $1-\rho = 2^{-k}$ - hold a constant ratio R = 2 at every k, using only $O(\log(1/(1-\rho\_max)))$ bins to cover the same range, versus $\Theta(1/(1-\rho\_max))$ equal-width bins for the same bound. This project's own calibration set reaches $\hat{\rho}$ up to 1.86 (deliberately oversaturated scenarios, Section III-A), well into the regime where this blowup is not a hypothetical edge case.

B. SA-Mondrian CP: A Sample-Size-Floored Fix

The theorem's natural remedy - geometric bins along $\hat{\rho}$ - fails in practice: tested directly against equal-width $\hat{\rho}$-binning on real calibration data, it loses at every bin count from 5 upward, collapsing to 50-57% worst-category coverage at 15 bins (versus equal-width's 80-84%). The reason is a finite-sample cost the asymptotic theorem does not see: geometric spacing concentrates bins in the sparse $\hat{\rho} \to 1$ tail, where this project's real scenario sweep has few calibration points, so empirical-quantile noise from tiny per-bin samples outweighs the scale-homogeneity benefit. We call the method that corrects this SA-Mondrian CP (Singularity-Anchored Mondrian CP): bins are built along $\hat{\rho}$ alone, starting from the highest-utilization calibration point and growing inward until each bin accumulates at least n_min points, directly targeting the finite-sample failure mode rather than bin geometry alone. n_min is itself selected from a 70/30 split of calibration data only - never the test set - mirroring exactly how ρ_cap is chosen for QT-CP (Section VI-B).

C. Results: A Real Loss to Mondrian CP, a Real Win Over QT-CP

A first look, on a single calibration/test split, made SA-Mondrian CP appear to match the 2-covariate Mondrian grid (two wins, two narrow losses). This project's own standing practice (Section IV) is not to trust a single split, so we repeated the comparison across 30 independent (calibration, test) draws - the same draws, and the same paired-significance discipline, used for this paper's central marginal-coverage result (Section V-C) - with SA-Mondrian's n_min reselected from calibration data alone on every repeat. The single-split tie did not hold up.

TABLE VI
WORST-CATEGORY COVERAGE, 30-REPEAT MEAN ± STD (TARGET 90%; P-VALUE IS SA-MONDRIAN VS. MONDRIAN, PAIRED T-TEST)

Target	Mondrian (2D)	QT-CP	SA-Mondrian (1D)	p-value
n_patients	85.3 ± 2.7%	71.3 ± 4.0%	83.1 ± 3.9%	0.009
mean_wait	84.3 ± 3.1%	80.3 ± 3.1%	83.7 ± 3.5%	0.404
mean_total	84.7 ± 2.7%	81.5 ± 3.2%	82.3 ± 4.8%	0.017
p95_wait	85.5 ± 2.9%	78.7 ± 4.3%	83.2 ± 2.4%	<0.001

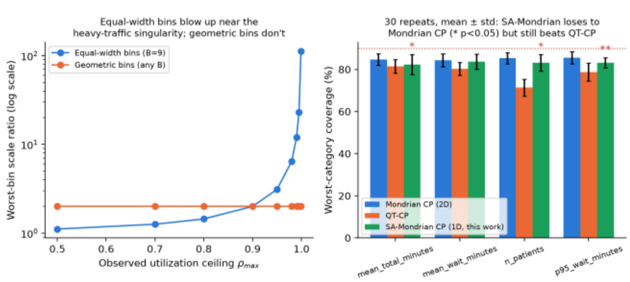

Fig. 12. Left: equal-width bins' worst-bin scale ratio vs. geometric bins', as a function of the observed utilization ceiling (log scale). Right: worst-category coverage, 30-repeat mean ± std, all four targets (* p<0.05, ** p<0.001, paired t-test vs. Mondrian CP).

SA-Mondrian CP loses to the 2-covariate Mondrian grid on every target - significantly on three of four (paired t-test, $p \leq 0.017$), with only mean_wait_minutes not reaching significance ($p = 0.40$). Collapsing Mondrian's two raw covariates into a single derived one costs real conditional-coverage performance: the 2D grid's fully nonparametric per-cell quantile evidently captures structure (interaction effects between staffing and arrival rate not reducible to their ratio $\hat{\rho}$ alone) that a 1D construction, however carefully binned, cannot. We report this as a genuine negative result for the hypothesis that $\hat{\rho}$ alone is sufficient, rather than reframing a loss as a tie. What does hold up under the same rigor, checked with the identical paired test: SA-Mondrian CP significantly outperforms QT-CP on three of four targets (mean_wait_minutes, n_patients, p95_wait_minutes; $p < 0.001$ each) - only mean_total_minutes shows no significant difference between them ($p = 0.44$). So while a single derived covariate is not enough to match Mondrian's own 2D partition, how that covariate's bins are built still matters more than QT-CP's fixed parametric rescaling of it, on most though not all targets.

D. A Natural Refinement That Changes Nothing

If replacing Mondrian's 2D grid with one derived covariate loses, a narrower question follows: does refining the 2D grid, rather than replacing it, help? Within each of Mondrian's 9 existing cells, $\hat{\rho}$ still varies somewhat, so we normalized residuals by QT-CP's $\hat{\sigma}(x)$ before calibrating each cell's own quantile - keeping Mondrian's partition intact and adding only a continuous within-cell correction. Tested with the same 30-repeat, paired-significance rigor, this makes no significant difference on any target ($|\Delta| \leq 0.54$ points, $p \geq 0.19$ throughout) - a genuine null result, not a small win reframed. The most likely reason: Mondrian's tercile-width cells are already narrow enough that $\hat{\rho}$, and so $\hat{\sigma}(x)$, barely varies within one - the normalization is close to a no-op. This is itself informative: whatever keeps Mondrian CP's own worst-category coverage at 84-86% rather than the full 90% target is not leftover within-cell utilization heterogeneity, and combining Mondrian's partition with QT-CP's rescaling - the direction Section VII-B might suggest - does not resolve it. We leave the actual source of that residual gap as an open question rather than speculate past what these results support.

VIII. DISCUSSION

The mechanism behind Section V-B's gap is a heteroscedastic, queueing-theoretic one, not an artifact of the surrogate or the CP procedure. Kingman's heavy-traffic approximation (Section VI-A) shows queueing delay - and its variance - diverging as utilization $\rho \to 1$, non-linearly and without bound. A single pooled nonconformity quantile - fit mostly to the many low-utilization scenarios in the calibration set - systematically undercovers the few high-utilization scenarios where this variance is largest. Mondrian CP's per-category quantile is exactly the correction this mechanism calls for: a separate quantile fit only to that category's own (larger) variance. This also explains why the gap is asymmetric across targets, and it is checkable directly against calibration-set residual variance per category (Table VII), not left as narrative: n_patients' per-category residual standard deviation barely spreads across the 9 categories and shows no significant conditional gap to correct, while wait-time targets' residual spread is far more regime-dependent, tracking directly with the size of each target's own conditional coverage gap.

TABLE VII
PER-CATEGORY CALIBRATION RESIDUAL STD., MIN-MAX (9 CATEGORIES)

Target	Min std	Max std	Max/min ratio
n_patients	5.1	9.3	1.8x
mean_total	2.8	11.5	4.1x
mean_wait	0.6	12.0	19.1x
p95_wait	4.8	115.6	23.9x

The practical implication for a staffing decision-maker is direct: a marginal coverage number, however statistically valid, is the wrong number to trust when the decision at hand is specifically about the high-utilization regime where marginal coverage is least informative about that regime's own reliability.

This mechanism can be stated precisely rather than only narratively. Suppose the nonconformity score at utilization ρ follows a location-scale model $S(\rho) =_d \sigma(\rho)\cdot Z$, $\sigma(\rho)$ increasing in ρ and Z a fixed-shape random variable - motivated by, though weaker than, the exact heavy-traffic limit Section VI-A cites.

$$Coverage(\rho) = P(S(\rho) \leq \hat{q}) = P(Z \leq \hat{q}/\sigma(\rho)) = F\_Z(\hat{q} / \sigma(\rho)) \quad (14)$$

Since $\sigma(\rho)$ is increasing, $\hat{q}/\sigma(\rho)$ is decreasing in ρ, and since F_Z is non-decreasing, Coverage(ρ) is a monotonically non-increasing function of ρ for any pooled quantile $\hat{q}$ - not merely observed empirically, but a necessary consequence of this model. We checked the model's own premise against real calibration data (mean_wait_minutes): fitting $\log(\text{scale}) \sim b\cdot\log(1/(1-\rho))$ gives $b \approx 0.92$, close to the $b = 1$ the heavy-traffic limit predicts. The exact closed-form miscoverage curve this implies, however - taking Z's limiting heavy-traffic shape (Exponential) literally - fits only loosely against the real per-bin miscoverage rate (RMSE $\approx$ 0.33 across ten ρ-bins): real degradation is more gradual than the pure asymptotic prediction, since most of the calibration sweep's mass sits well below the $\rho \to 1$ limit where that exact shape holds. We report the monotonicity result as structural and load-bearing, and the specific closed-form curve as a loose approximation outside heavy traffic, not a validated quantitative law - an honest distinction we think matters more than a tidier but overstated claim would.

IX. LIMITATIONS AND THREATS TO VALIDITY

Service-time distributions are literature-calibrated log-normals, not recoverable from the source dataset - disclosed explicitly throughout rather than blended with the real-data-derived arrival and acuity distributions, but a real gap relative to a fully real-data-calibrated DES. Cross-site replication (Section V-D) covers two departments from one health system; a third, structurally distinct site (e.g., a different health system or country) would strengthen the generalizability claim further. The exchangeability-violation extensions (CRC, ACI, weighted CP) are each evaluated on a single stress-test design rather than the 30-repeat significance testing applied to the core Mondrian result in Section V-A - appropriate given this paper's scope, but a narrower evidentiary standard than the paper's central finding, and 5 of the architecture-benchmark results in Section V-F are summarized rather than reported in full for space. Finally, the operational conclusion drawn (Section VIII) - that marginal coverage is the wrong number for a high-utilization staffing decision - is drawn from a simulated environment; production deployment against live patient data was intentionally out of scope.

X. CONCLUSION

We test both explicit limitations [3] leave open for conformal prediction on surrogate models - a marginal-only guarantee and an untested exchangeability assumption - in a discrete-event queueing domain outside the physics simulations they study. Standard CP's marginal guarantee conceals a severe, 22-point conditional coverage failure in exactly the highest-stakes operating regime; Mondrian CP restores coverage there without sacrificing the marginal guarantee, a result confirmed significant across repeated trials and replicated at an independent site. Under severe exchangeability violation, both methods' advantage collapses together, and we quantify how much three principled corrections recover. We also propose QT-CP, a new bin-free nonconformity score grounded in Kingman's heavy-traffic approximation, and report it honestly: a genuine width-efficiency gain over standard CP, but not a replacement for Mondrian CP's conditional-coverage repair. Examining why motivates a further result: equal-width bins along the utilization axis have provably unbounded scale distortion as utilization approaches 1, and the geometric-spacing fix this implies fails empirically from calibration-sample scarcity in the tail - correcting for which yields SA-Mondrian CP, a sample-size-floored adaptive binning scheme along that one derived covariate. Tested with the same 30-repeat rigor as this paper's central finding rather than trusted from a single split, it significantly underperforms the 2-covariate Mondrian grid on three of four targets - a genuine negative result for the hypothesis that a single derived covariate suffices - while still significantly outperforming QT-CP on three of four, evidence that how bins are built along the difficulty axis matters even where how many covariates define it does not fully substitute for Mondrian's own nonparametric partition. The natural next attempt - refining Mondrian's own grid with QT-CP's continuous rescaling rather than replacing it - changes nothing (a genuine null result, Section VII-D), narrowing rather than resolving the open question of what keeps even Mondrian CP's own worst-category coverage below its 90% target. These results argue that, in queueing-driven operational domains specifically, conditional - not only marginal - coverage should be the reported and trusted quantity wherever the decision itself is conditional on the operating regime.

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